

Cisco IoT Design-In Program Guide

Introduction

Organizations everywhere are digitizing to solve business problems, reduce costs, increase growth, improve the customer experience, and optimize operations. Digital transformation is real, and companies without a strategy will fall behind. Thus, organizations are increasingly engaging in Internet of Things (IoT) initiatives.

Cisco has extensive experience providing integrated solutions – whether they are boards embedded in a product or a stand-alone router or other network device that is included as part of a larger solution. We have been working with integrators, industrial control systems vendors, and other equipment builders for years, helping to connect and implement IoT into their solutions.

Our experience and capabilities include connecting “things” (machines, sensors, workforces), from support for a vast array of industrial protocols, secure network segmentation and configuration, management and remote monitoring, and network security, to delivering data-driven business outcomes (eliminating downtime, driving predictive maintenance, enabling asset visualization, increasing productivity, and streamlining operations).

Designing, deploying, and selling an IoT solution can be a daunting task full of complexities and unforeseen obstacles, causing 60 percent of IoT initiatives to stall at the proof of concept stage. The challenge begins with the fact that every IoT integration is different. Every customer has different needs driving an unlimited number of use cases and an unlimited number of variables. Deciding which product to use and how to configure and deploy it can be overwhelming – especially as experts are few and far between. These deployments involve complex infrastructure and operational processes. It is very hard to repeat an implementation due to the complex nature of technology and lack of standardized approach. Each solution requires a multitude of partners to manage and coordinate the project. Cost overruns, long lead times, and frustrated customers are the norm. And worse, the complexity impacts your revenue and your profitability.

Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

The Cisco® IoT Design-In Program

Standardize, Simplify, and Accelerate

Introducing the [Cisco® IoT Design-In Program](#), the industry's first-ever systematic lifecycle program for industrial networking. It simplifies and accelerates IoT projects by discovering, documenting, and using best practices to scale. The Design-In initiative standardizes the integration of Cisco's IoT product portfolio and support services into your equipment or solutions. It **speeds time to market, improves profitability, and increases reliability** through reference guides, learning labs, developer sandboxes, and Cisco customer support. These services are the synthesis of Cisco's decades of experience of working with integrators, Industrial Control System (ICS) vendors, and equipment builders.

The program provides commercialization services to qualified partners, including a commercial agreement tailored to fit your needs, authorization to use specific Cisco logos, and co-marketing activities. The program authorizes approved partners to resell Cisco products and support, either embedded into products or integrated within their solution.

What can you do with it?

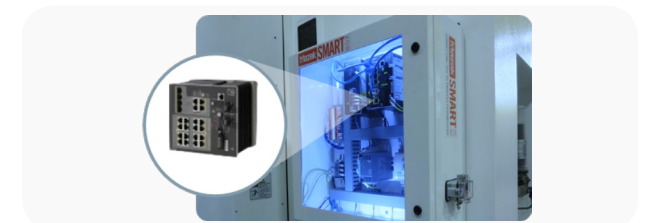
There are two methods to incorporate Cisco Design-In into your offerings:

- **Product Design-In:** Cisco networking boards embedded into purpose-built appliances or equipment. For example, the Cisco Embedded Services 3300 Series Switches (ESS 3300) and the Cisco ESR6300 Embedded Series Routers (ESR 6300) are used by systems integrators to add secure networking to a wide range of defense equipment such as vehicles, tanks, and drones.
- **Solution Design-In:** The Cisco IoT product portfolio is ruggedized for a wide range of industrial specifications, such as temperature, vibration, and dust. These off-the-shelf industrial products are often integrated into a wide range of industry solutions.

Figure 1. ES S 3300 and ESR 6300



Figure 2. Mazak SmartBox



Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Who is this for?

The Cisco IoT Design-In Program is intended for a wide range of partners, ranging from IT professionals working with their OT counterparts to OT professionals working at systems integrators, ICS vendors, and equipment builders.

The affected roles include everyone from operators on the shop floor (OT), controls engineers, and technicians to plant managers (OT) and COOs. These professionals are ultimately focused on delivering operational efficiency and business outcomes.

The program is also relevant for IT professionals who are working with their OT counterparts. These include system architects network administrators, developers, and Chief Security Officers (CSO) who together ensure that the network infrastructure and applications perform optimally and securely.

A three-step solution

Design-In follows a three-step process to accelerate your IoT journey:

1. Discover

The first step is to help you identify the Cisco product for your needs. You can view the standardized reference guides to find the one that best fits your environment. The use case catalog links to all the tools and resources to help you design and plan your deployment. We also offer a self-guided assessment and/or workshop to qualified partners to help jumpstart this process.

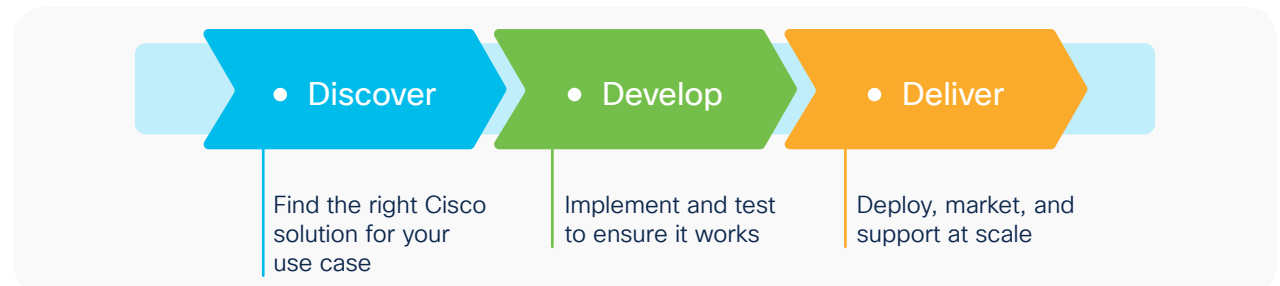
2. Develop

You can reduce time spent figuring out the integration and increase project reliability and success. Our award-winning developer program, DevNet, gives you all you need to develop and test your solution.

3. Deliver

We offer step-by-step guidance to become a Cisco partner with rights to resell and sales enablement support, plus world-class access to Cisco marketing resources, the right to use specific Cisco logos, and global technical support specifically designed for the needs of industrial partners.

Figure 3. Cisco IoT Design-In three-step summary



Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Step 1: Discover

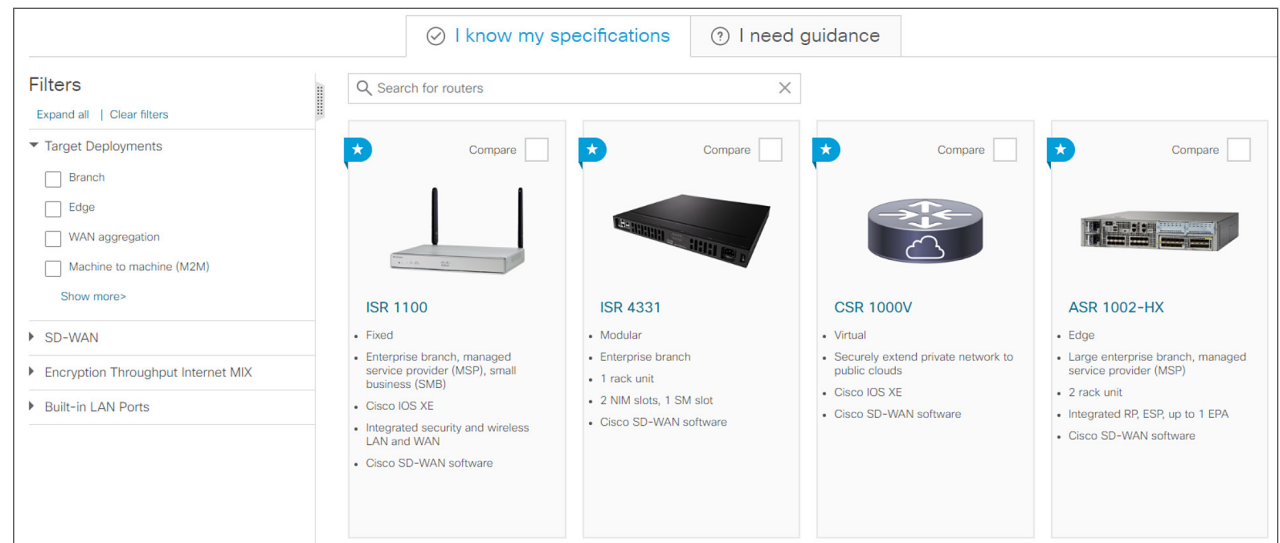
Roles involved: Control systems engineer, line-of-business manager, solutions architect, network/software engineer

The first step of Design-In is **Discover**. This phase streamlines your up-front project scoping through special tools and resources, allowing you to create a standardized design and solution.

Start your journey to the right solution by finding a use case and/or product that best fits your current situation and requirements. This is accomplished with Cisco's Design-In IoT [Product Selector](#) and IoT Use Case Selector tools.

Do you have a general understanding of the networking hardware you need? Maybe you know you need a router or a managed switch, but you don't know what's available or you haven't determined what all of the specs should be. Cisco's IoT Product Selector tool can help with that. The Product Selector allows you to choose the best Cisco IoT product through filter-based suggestions similar to most e-commerce shopping websites. Once you have chosen the filters that match your requirements, you will be shown the products that meet your criteria, and you can compare them side by side.

Figure 4. Cisco IoT Product Selector



Contents

Introduction

- The Cisco® IoT Design-In Program
- What can you do with it?
- Who is this for?
- A three-step solution

Step 1: Discover

- Professional Services workshops

Step 2: Develop

- Design-In reference guides
- Sandboxes and learning labs

Step 3: Deliver

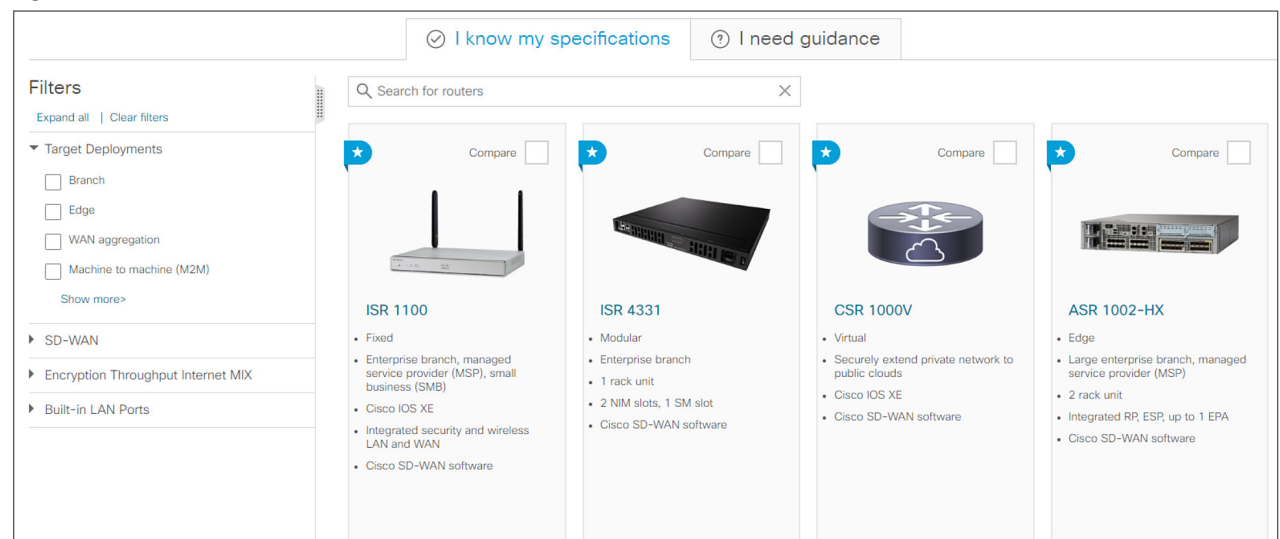
- Commercial resale agreement
- Technical support

Becoming a Cisco partner

Summary

Need to connect a remote machine without network connectivity using a gateway? There's a use case for that. Aiming to use industrial switching securely in a factory environment? There's a use case for that. These (and other scenarios) are common examples in industrial environments, which Cisco has documented for you to use as a baseline, or purely as inspiration. Determine which of these use cases works best for you by using the Cisco IoT Use Case Selector tool. The Use Case Selector guides you through a question-based assessment, which determines the best fit based on your responses. Once you have finished the assessment, your options will be shown, and you can compare them side by side.

Figure 5. Cisco IoT Use Case Selector



Upon selecting your product or solution, we help you go a level deeper, with best practices that help you implement the solution that best fits your needs.

The Use Case Catalog simplifies planning, as each use case is based on actual deployments. The catalog provides best practices, deployment instructions, and implementation guidelines. Each use case has also been tested in our labs to help ensure that your standardized design complies with our best practices.

As an additional element of **Discover**, Cisco offers the opportunity to run a workshop with your product and business teams to complete this phase collaboratively. From hands-off to hands-on, Discover is a mix of self-assessment with optional deeper engagement.

Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Professional Services workshops

For customers that require deeper guidance, call upon the Cisco Professional Services Group. With packages ranging from three days to customized workshops available on several key topics, including cybersecurity and edge computing, these workshops will teach you about cutting-edge networking technology and best practices. Knowing how others implement this technology today for tomorrow will help your team stay ahead of the competition.

Step 2: Develop

Roles involved: Controls system engineer, operator, technician, developer

After determining which Cisco products and use cases best fit your needs, the next step is deciding how to implement the chosen use case. You can get a hands-on experience of select use cases to ensure that you have a firm understanding of the technologies before developing and deploying the solutions that you create and deliver into production. Some of the use cases are accompanied by learning labs and sandboxes that are available for hands-on training in simulated environments. You can leverage the industry's most complete Design-In Reference Guides that provide step-by-step planning and implementation help for your use cases. These guides are made available to speed integration, all supported by Cisco's industry-leading developer program, [DevNet](#). You can rely on a community of thousands of developers to help you go faster. You can help ensure success with integration validation testing (IVT), in which you authorize a third party to validate your production code and confirm that it will both successfully run and comply with Cisco systems configurations.

Design-In reference guides

A Design-In reference guide is a description of a reference customer's environment, key challenges, and solution goals, and a step-by-step implementation guide. These guides are based on Cisco's experience with real-world, proven, and repeatable implementations and are tested in our labs.

Currently, the Design-In reference Guides are broadly divided by product lines and the most common use cases.

Industrial Ethernet switches

- Industrial Networking Reference Guides with Industrial Ethernet switches
 - [Connected Machines Reference Guide](#)
 - [Managed Networks and Cyber Security Reference Guide](#)
 - [Industrial Compute Reference Guide](#)

Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Embedded products

- Embedded products
 - [Embedded Design Patterns Reference Guide](#)

Connecting Machines Remotely

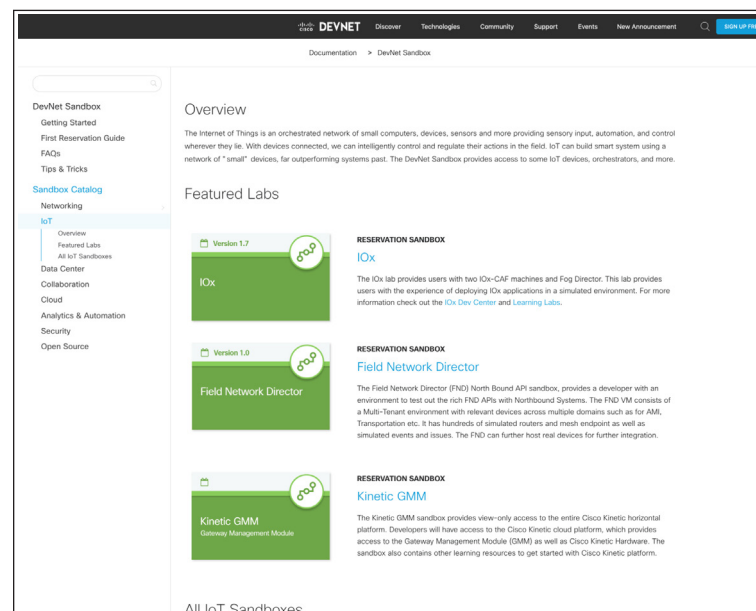
- [Remotely Connecting Machines Reference Guide](#)

Sandboxes and learning labs

To make your learning experience more effective, we provide you with sandboxes that mirror the deployment characteristics of your use cases. Considering the safety and production impact risk of our technologies, experimenting with live systems is not usually an option. Sandboxes provide you with a safe and isolated environment to test and learn our products.

Once you have signed up with DevNet and have reserved a sandbox instance, you will find that each sandbox is associated with multiple learning labs that you can follow at your own pace. The suggested order for completing the learning labs mirrors our step-by-step reference design and implementation guides.

Figure 6. Cisco IoT DevNet sandboxes



Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

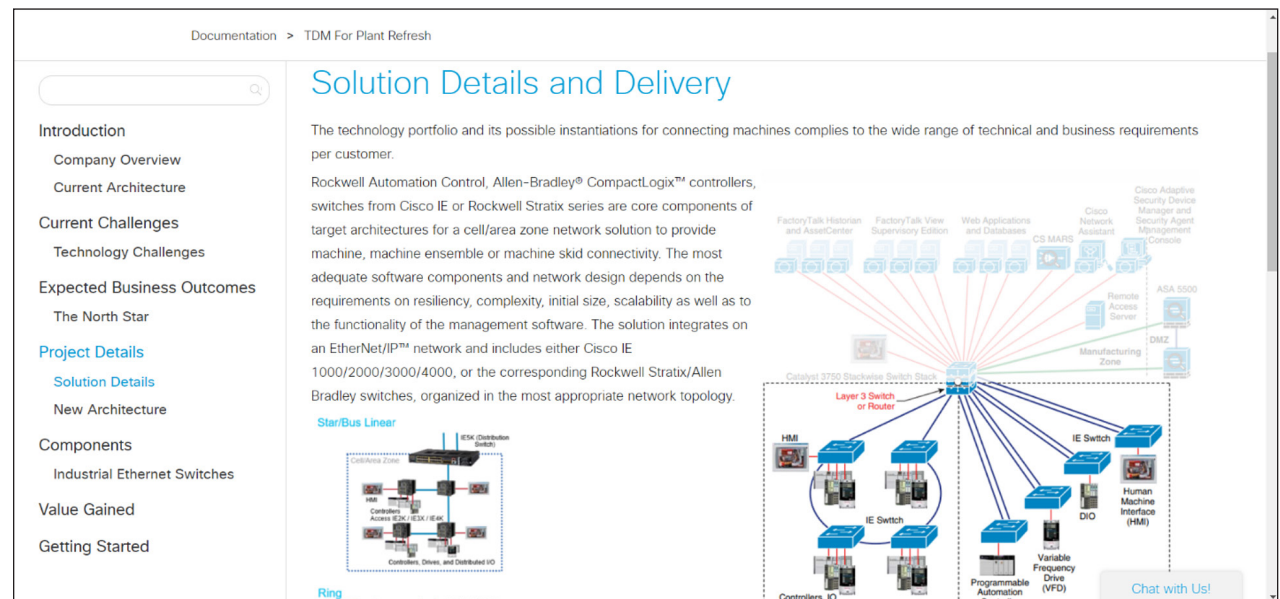
Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Your design journey begins with reading the step-by-step guides based on the use cases, builds up with experimenting with our technologies in simulated environments, and continues with workshops. Our team of experts visits you at your work location or you can travel to one of our technology centers for an in-person, hands-on learning experience with our products – again based on your specific use case.

Figure 7. Cisco IoT reference guide



Going forward, the same team of experts stays available to you throughout your technology adoption journey, whenever you need help and best-practice guidance.

Contents

Introduction

- The Cisco® IoT Design-In Program
- What can you do with it?
- Who is this for?
- A three-step solution

Step 1: Discover

- Professional Services workshops

Step 2: Develop

- Design-In reference guides
- Sandboxes and learning labs

Step 3: Deliver

- Commercial resale agreement
- Technical support

Becoming a Cisco partner

Summary

Step 3: Deliver

Roles involved: Legal, finance, operations, Customer Experience (CX) TAC support, marketing, product management

In the final phase of the Design-In process we have **Deliver**, in which we help speed your go-to-market. From commercial agreements to marketing to technical support, we help you sell to and support your customers efficiently and effectively.

Our new commercial agreements have a special discount structure and are simplified to meet your needs. They include the ability to resell our products and technical support for our products used by industrial control system vendors, equipment builders, and integrators. We provide Level 1 and Level 2 Cisco product support training to our partners, so that you can provide fast and efficient customer support to your end customers who are purchasing your product or solution. Cisco will stand behind you for product issues not resolved by the Level 1 and Level 2 support.

As part of the program, Cisco authorizes approved partners to use select Cisco logos on either their shipping products or in their co-marketing activities such as at events, seminars, and collateral. The widely recognized Cisco logo gives your customers the confidence that your product has Cisco security and quality built in.

Cisco could also invite select partners to participate in co-marketing activities, such as speaking opportunities at industry events, marketing campaigns, and joint collateral.

Commercial resale agreement

Cisco will work with you to select the appropriate commercial channel agreements that best meet your needs. These channel agreements authorize partners to purchase and resell Cisco products and technical support through approved distributors.

Cisco understands that the partners' products and solutions (that incorporate Cisco technologies) are sold through a wide range of industrial channels: direct, standard distributor, Value-Added Resellers (VARs), and via OEM or embedded licenses. Our agreements have been designed with these channels in mind. For instance, we may not require you to Report Point-of-Sale (POS) customers if you don't have visibility into them (as is the case with an OEM). The agreements have also been simplified and streamlined to the most basic terms and conditions essential to protect both you and Cisco, and to comply with essential regulations.

Cisco has also looked at market metrics that provide the most attractive built-in discounts. In addition, Cisco is very clear about key restrictions and requirements for the discounts, particularly that the Cisco products be sold together with your intended/stated products and not separately.

Contents

Introduction

- The Cisco® IoT Design-In Program
- What can you do with it?
- Who is this for?
- A three-step solution

Step 1: Discover

- Professional Services workshops

Step 2: Develop

- Design-In reference guides
- Sandboxes and learning labs

Step 3: Deliver

- Commercial resale agreement
- Technical support

Becoming a Cisco partner

Summary

Technical support

You can create a new revenue stream and ensure that your customers have the technical support they need with Cisco’s technical support service. Key benefits include:

- Reduce downtime with 24-hour access to Cisco’s Technical Assistance Center (TAC)
- Stay up to date with access to software updates, patches, and alerts
- Have clear visibility into your contract coverage and support usage with coverage status and usage reports

Additional options are available for partners who want services such as next-business-day hardware replacement and software application support, and other add-ons.

Cisco offers two different support packages: Cisco [Smart Net Total Care® Service](#) and the Industrial Technical Support program. Both are tailored to resellers, but Industrial Technical Support is specifically designed for partners under the Cisco IoT Design-In channel program agreement.

Table 1. Features of support packages

	Smart Net Total Care	Industrial Technical Support
Contract ownership and management	End customer	Partner
Service offering	Cisco branded service (Cisco to end customer)	Partner-branded service (Cisco to partner)
Operational support relationship	End customer contacts Cisco	Partner contacts Cisco
Support responsibilities	Level 2 and 3 support by Cisco	Level 1 and 2 support by partner
Asset contract information	End-customer name and installed-at information	Partner name and office/spare drop-off information

Contents

Introduction

The Cisco® IoT Design-In Program
What can you do with it?
Who is this for?
A three-step solution

Step 1: Discover

Professional Services workshops

Step 2: Develop

Design-In reference guides
Sandboxes and learning labs

Step 3: Deliver

Commercial resale agreement
Technical support

Becoming a Cisco partner

Summary

Becoming a Cisco partner

In order to execute the Design-In channel partner agreement, you must become a Cisco Partner. Cisco has decades of experience in working with channel partners, helping them maximize growth and provide differentiation and profitability by extending their capabilities to current and new markets. The IoT Design-In program in the Deliver phase authorizes partners to resell Cisco technology embedded or integrated directly into their products or within a larger solution through their channels. It provides industrial partners with commercial benefits and supply chain process support needed to scale solution delivery.

Summary

The [Cisco IoT Design-In Program](#) is the industry's first-ever systematic lifecycle program for industrial networking. It simplifies and accelerates IoT projects by discovering, documenting, and using best practices to scale. Design-In accelerates your IoT journey through three simple steps:

1. Discover: Streamline your up-front project scoping through special tools and resources, allowing you to create a standardized design and solution.
2. Develop: Reduce time spent figuring out the integration and increase project reliability and success with our award-winning developer program, [DevNet](#).
3. Deliver: Speed time to revenue with Cisco by deploying and supporting at scale.

The program provides you with step-by-step deployment guides, preintegrated and validated designs, customer support, tools and trainings, and go-to-market support, increasing the reliability and success rate of IoT projects.